INTEGRATED HEALTH MONITORING SYSTEM FOR BLOOD PRESSURE, BLOOD GLUCOSE, AND BODY TEMPERATURE

M1:AHMED ALQARNI | M2: AHMED ALAHMADI | M3: SADATI MWASONGWE | M4: BASSEL ALNAJEM

Department of Electrical and Computer Engineering Biomedical Engineering Program

Supervisor: DR. AHMAD TURKI

INTROUDCTION

This project focuses on the development of a wearable device designed to monitor critical health parameters, including blood pressure, body temperature, and blood glucose levels, in real-time. The device integrates multiple sensors: MAX30102 (for PPG), AD8232 (for ECG), DS18B20 (for body temperature), and BME688 (for blood glucose approximation via acetone detection). Utilizing the Pulse Transit Time (PTT) algorithm, blood pressure is calculated non-invasively, while body temperature is displayed in both Celsius and Fahrenheit. The system is equipped with IoT functionality for mobile notifications, ensuring continuous health tracking. This wearable aims to provide a compact, non-invasive, and cost-effective solution for personalized health monitoring, enhancing convenience and accessibility for users.

BLOCK DIAGRAM & CIRCUIT SCHEMATIC

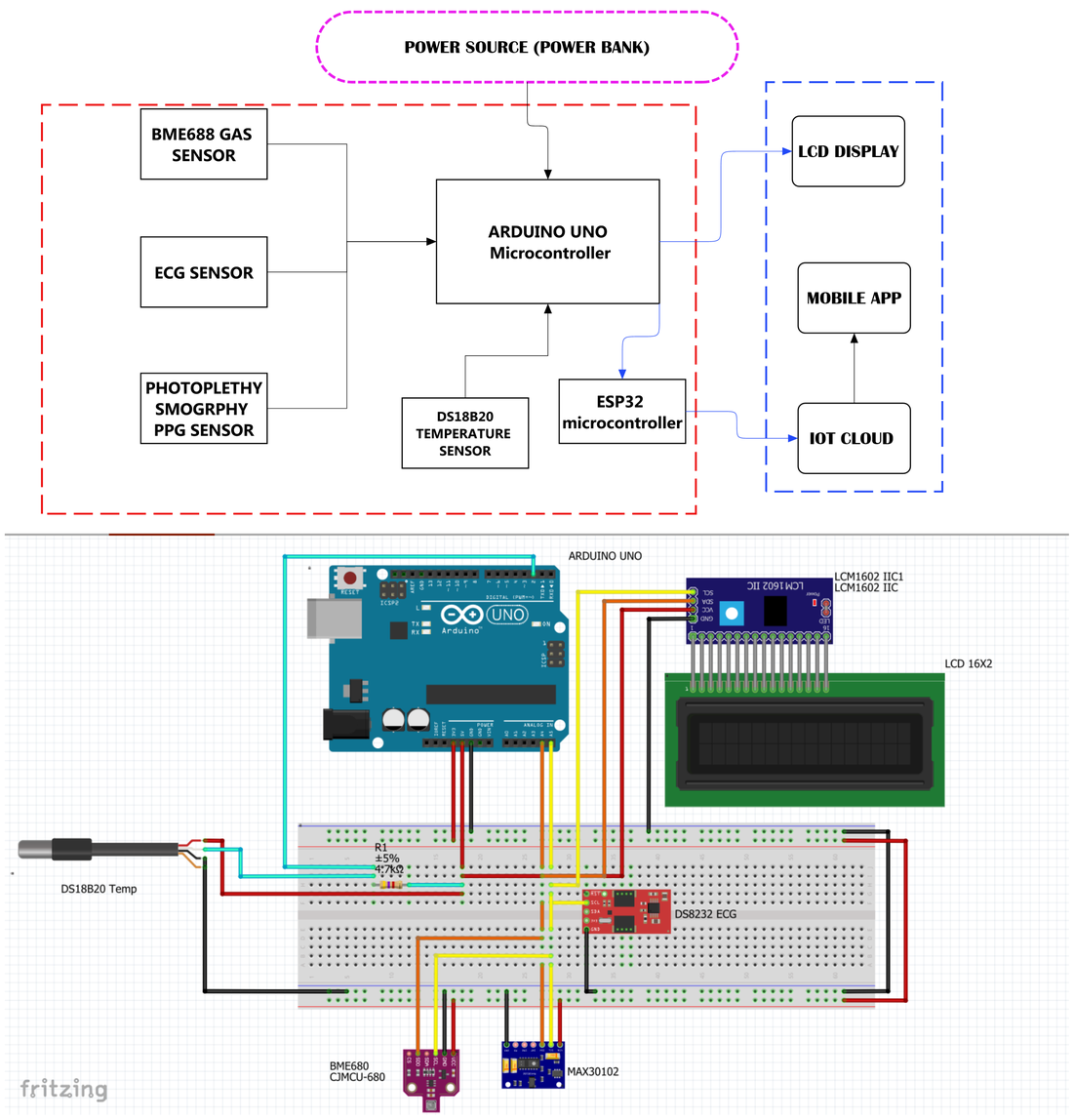


Fig.1 the block diagram of our baseline design and the schematic circuit diagram

COMPONENTS SPECIFICATIONS

Component	Model	Function	Key Specifications
Gas Sensor	BME688	Measures breath acetone (proxy for glucose)	High sensitivity, I2C interface
Pulse Oximeter Sensor	MAX30102	Measures PPG for blood pressure calculation	Low power, integrated LEDs and photodetector
ECG Module	AD8232	Measures ECG signals for blood pressure	Low noise, compact size
Temperature Sensor	DS18B20	Measures body temperature	Accuracy ±0.5°C, digital output
Microcontroller	Arduino UNO R3	Controls and processes all sensor data	ATmega328P, 6 analog input pins
Display	I2C LCD	Displays measurement results	16x2 characters, I2C interface
Power Supply	Power Bank/Battery	Powers the entire wearable system	5V output, rechargeable

IMPLEMENTATION & VALIDATION

- **Blood Pressure Measurement**:
  - Sensors: MAX30102 (PPG) and AD8232 (ECG).
  - Method: Pulse Transit Time (PTT) algorithm.
  - Output: Systolic and diastolic values on an I2C LCD.
- **Body Temperature Measurement**:
  - Sensor: DS18B20.
  - Output: Temperature in Celsius and Fahrenheit on an I2C LCD.
- **Blood Glucose Approximation**:
  - Sensor: Bosch BME688 (acetone detection).
  - Method: Regression model for glucose estimation.
- **Integration**:
  - Microcontroller: Arduino UNO.
  - Real-time display and IoT notifications for remote monitoring.

Design calculations

Diastolic Blood Pressure (DBP):

$$\mathrm{DBP} = 0.015 \times \mathrm{PTT} + 78.799$$

Systolic Blood Pressure (SBP):

$$\mathrm{SBP} = -0.024 \times \mathrm{PTT} + 132.861$$

Pulse Transit Time (PTT) Calculation (From ECG and PPG):

$$\mathrm{PTT} = \frac{t_{\text{R-peak (ECG)}} - t_{\text{PPG peak}}}{\text{sampling rate}}$$

Celsius (°C) = Direct reading from DS18B20 sensor.

Fahrenheit (°F):

$$^{\circ}F = (^{\circ}C \times \frac{9}{5}) + 32$$

Concentration of Gas (Acetone) in ppm (Parts per Million):

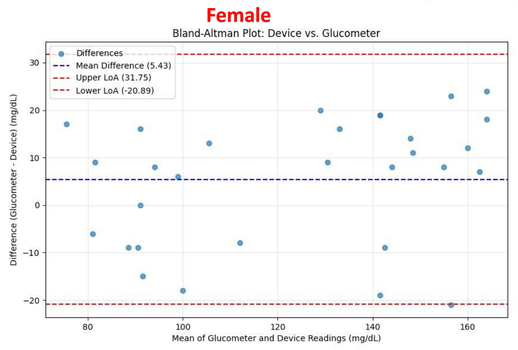
$$\text{Gas Concentration (ppm)} = k \times \left(\frac{R_{\text{sensor}}}{R_{\text{clean air}}}\right)$$

$$\text{Blood Glucose (mg/dL)} = a \times \text{Acetone Concentration (ppm)} + b$$

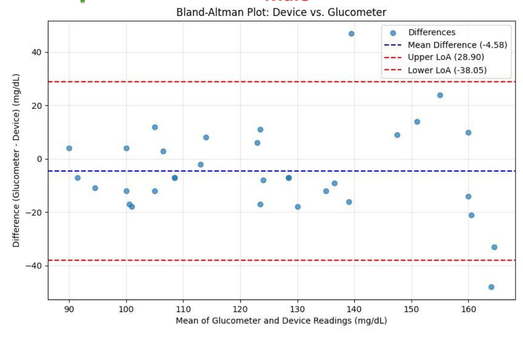
Blood Glucose Validation

The Bland-Altman plots

Female



Male



Participant ID	Age	Height(m)	Weight(KG)	BMI	Trial Number	Glucometer Reading (mg/dL)	Device Reading (mg/dL)	Difference (mg/dL)	Eroor (%)
1	20	1.58	47	18.8	1	98	90	8	8.16
					2	176	152	24	13.64
					3	138	147	-9	-6.52
2	23	1.68	65	23.0	1	84	93	-9	-10.71
					2	166	159	7	4.22
					3	118	111	7	5.93
3	30	1.62	53	20.2	1	99	91	8	8.08
					2	146	155	-9	-6.16
					3	108	116	-8	-7.41
4	24	1.70	57	19.7	1	86	95	-9	-10.47
					2	135	126	9	6.67
					3	112	99	13	11.61
5	32	1.59	68	26.9	1	102	96	6	5.88
					2	166	154	12	7.23
					3	139	119	20	14.39
6	28	1.71	69	23.6	1	91	109	-18	-19.78
					2	155	168	-13	-8.39
					3	141	125	16	11.35
7	38	1.66	58	21.0	1	84	99	-15	-17.86
					2	145	168	-23	-15.86
					3	124	138	-14	-11.29
8	22	1.72	76	25.7	1	106	90	16	15.09
					2	173	155	18	10.40
					3	148	155	-7	-4.73
9	19	1.55	58	24.1	1	78	84	-6	-7.69
					2	146	167	-21	-14.38
					3	132	151	-19	-14.39
10	24	1.57	60	24.3	1	99	83	16	16.16
					2	161	149	12	7.45
					3	151	132	19	12.58
								Error average	10.48

Participant ID	Age	Height(m)	Weight(KG)	BMI	Trial Number	Glucometer Reading (mg/dL)	Device Reading (mg/dL)	Difference (mg/dL)	Eroor (%)
1	20	1.68	91	32.2	1	88	95	-7	-7.95
					2	153	167	-14	-9.15
					3	126	120	6	4.76
2	27	1.75	82	26.8	1	111	99	12	10.81
					2	158	149	9	5.70
					3	129	141	-12	-9.30
3	26	1.77	70	22.3	1	98	116	-18	-18.37
					2	163	182	-19	-11.66
					3	123	117	6	4.88
4	23	1.69	68	23.8	1	92	109	-17	-18.48
					2	140	88	52	37.14
					3	121	139	-18	-14.88
5	21	1.71	74	25.3	1	89	99	-10	-11.24
					2	148	181	-33	-22.30
					3	115	132	-17	-14.78
6	24	1.83	69	20.6	1	112	114	-2	-1.79
					2	120	128	-8	-6.67
					3	108	105	3	2.78
7	24	1.78	58	18.3	1	92	88	4	4.35
					2	167	143	24	14.37
					3	105	112	-7	-6.67
8	20	1.64	55	20.4	1	102	98	4	3.92
					2	165	155	10	6.06
					3	132	141	-9	-6.82
9	22	1.72	63	21.3	1	99	111	-12	-12.12
					2	150	171	-21	-14.00
					3	131	147	-16	-12.21
10	22	1.75	68	22.2	1	94	106	-12	-12.77
					2	148	138	10	6.76
					3	118	110	8	6.78
11	21	1.85	72	21.0	1	90	98	-8	-8.89
					2	152	143	9	5.92
					3	125	132	-7	-5.60
								Error average	10.30

CONCLUSION

The wearable device integrates the BME688, MAX30102, and DS18B20 sensors to non-invasively measure blood glucose, blood pressure, and body temperature in real time. Utilizing advanced algorithms for signal processing, it delivers accurate and reliable results, validated through Bland-Altman analysis against standard measurements for both male and female participants. Its compact, cost-effective, and user-friendly design ensures easy operation, making it an ideal solution for continuous health monitoring and early detection of health anomalies across diverse populations.

The final design



Fig.2. The complete device is shown on the right, with a hand strap for PPG measurement on the left.